

Activity 0.1 – Everyday Task as an Algorithm

Title: From Daily Routine to Algorithm

Instructions to Students

1. Choose one simple daily activity (examples: making tea, preparing to go to class, booking a cab).
2. Write the steps in clear, numbered points.
3. Make sure each step is:
 - Unambiguous
 - In correct order
 - Does not skip important details

Template

1. Activity chosen: Making Maggie

2. Write the steps:

1. Boil 150 ml of water per maggie in a bowl
2. after it comes to a boil, add the maggie.
3. as it reaches softness, add the masala packet
4. Cook the maggie for 5 more minutes
5. Top it with a pinch of salt and garnish with coriander
6. Serve it hot in a plate. or a bowl.

3. Check your algorithm:

- Can a friend follow it without asking questions? Yes / No
- If "No", refine the steps here:

• _____

Activity 0.2 – From Steps to Pseudo-code

Title: Writing Pseudo-code with IF and REPEAT

Instructions

1. Take your steps from Activity 0.1.
2. Identify where decisions happen (IF/ELSE).
3. Identify where repetition happens (REPEAT / WHILE).
4. Rewrite your algorithm using the following keywords:
 - IF, ELSE, ELSE IF
 - REPEAT, WHILE, UNTIL

Template

1. Original activity: _____
2. Pseudo-code version: _____

text

START

Take maggie & water and bring it to boil
IF there is > 2 packets (soft). THEN
use 1.5 cups of water additional to each packet (add masala)
ELSE
use 1.5 cup of water (wait till soft).
ENDIF

REPEAT

boiling at medium-high flame (check regularly for softness)
UNTIL fully cooked. (soft).

END

3. Reflection:

- Where did you use IF/ELSE? _____
 - Where did you use REPEAT/WHILE? _____
-

Activity 1.1 – Run and Edit Given Code

Title: First Python Script – Edit and Run

Instructions

1. Your teacher will give you a small Python script (example below).
2. Do NOT write new code first; only change text or numbers.
3. Run the program and observe the output.

Example Script

python

name = "Student"

age = 18

print("Hello," , name)

print("You are", age, "years old.")

Tasks

1. Change name and age to your own values.

- name: "thanishk"
- age: 21

2. Run the program. What is the output?

text

Output:

Hello, . thanishk
You are 21 years old

3. Change the printed message to:

- "Welcome to Python, <your name>"

Write your edited code here:

python

Edited code:

name = "thanishk"

~~age~~ print("welcome to python," , name)

4. Reflection:

- What changed in the output and why?

print statement changed

Activity 1.2 – Predict–Then–Run

Title: Predict Program Output

Instructions

1. Look at each code snippet.
2. BEFORE running, write what you think the output will be.
3. Then run it and compare.

Snippet A

python

x = 5

y = 2

z = x + y * 3

print(z)

- Predicted output: 11
- Actual output: 11

Snippet B

python

a = 10

b = a - 3

a = a + 1

print(a, b)

- Predicted output: 11, 7
- Actual output: 11 7

Reflection

- Where was your prediction wrong? Why?

Activity 2.1 – Calculator in Steps

Title: From Word Problem to Code

Problem 1

A student has marks in 3 subjects. Find total and average.

1. Write the steps in plain language:

1. total = sum of all the marks scored in 3 subjects
2. to find average, first take the total sum of marks
3. then divide that total with the number of subjects
4. the value derived after division is the average marks of all three subjects

2. Now convert to Python (fill in the blanks):

python

mark1 = 75

mark2 = 25

mark3 = 100

total = ~~xxx~~ mark1 + mark2 + mark3

average = ~~xxx~~ total / 3

print("Total =", total)

print("Average =", average)

3. Test with your own numbers and record output.

Activity 2.2 – Fix the Errors

Title: Debugging Simple Python

Instructions

Each snippet has an error.

1. Write what you think is wrong.
2. Correct the code.

Snippet 1

python

print("Hello World)

- Error: there is no double inverted commas at the end of statement)
- Corrected code: print("Hello World")

python

undefined

Snippet 2

python

a = 10

b = "5"

c = a + b

print(c)

- Error: 5 is an integer, using inverted comma will give error
- Corrected code:

python

undefined

Reflection:

```
a = 10
b = 5
c = a + b
print(c)
```

- What did you learn about syntax vs logic errors?
-
-

Activity 3.1 – Rule-based Classification

Title: Grades with IF/ELSE

Rule

- If mark ≥ 90 : Grade A
- Else if mark ≥ 75 : Grade B
- Else : Grade C

Tasks

1. Draw the decision in simple text:

- Condition 1: If marks is 90 or more than 90, then grade A.
- Condition 2: If marks is 75 or more, then grade B

2. Now write Python code: If marks less than 75, then grade C

python

```
mark = int(input("Enter mark: "))
```

```
if mark >= 90:
```

```
    print("Grade A")
```

```
elif mark >= 75:
```

```
    print("Grade B")
```

```
else:
```

```
    print("Grade C")
```

3. Test with at least 3 marks and note outputs.

Activity 3.2 – Mini Problem Bank

Title: Structured Problem Solving with IF/ELSE

For each question, fill the template.

Template

1. Restate the problem in one sentence:

Check the number if it is even or odd.

2. Conditions needed (bullets):

- If the number is divisible by 2
- If the number is not divisible by 2

3. Python code: $n = \text{int}(\text{input}(\text{"enter a no."}))$

python

```
if n % 2 == 0:  
    print("even")  
else:  
    print("odd")
```

Your solution

4. Test cases and results:

- Input: 4 Output: even
- Input: 9 Output: odd

Use this template for:

- ✓ Odd or even number.
- ✓ Ticket discount if age < 12 or > 60.
- Check if a year is a leap year (simple rule).

1. There is discount for tickets for people below 12 yrs age or above 60 yrs.

2. If age is below 12 / If age is above 60.

3. $\text{age} = \text{int}(\text{input}(\text{"enter age : "}))$

```
if age <= 12:  
    print("discount price for ticket")
```

```
elif age >= 60:  
    print("discount price for ticket")
```

```
else:  
    print("full price for ticket")
```

$\text{days} = \text{int}(\text{input}(\text{"no. of days in the year : "}))$

```
if days == 366:  
    print("this year is a leap year")
```

```
else:  
    print("not a leap year")
```

if age <= 12 or age >= 60:

else:

4. i/p = 11 o/p = discount

i/p = 25 o/p = full price

5. $\text{if } 12 > \text{age} > 60:$
 $\text{print}(\text{"discount"})$

else:
 $\text{print}(\text{"no discount"})$

$\text{years} = \text{int}(\text{input}(\text{"year : "}))$

```
if year % 4 == 0:  
    print("leap year")
```

```
else:  
    print("not a leap yr")
```


Activity 4.1 – Sum with Loops

Title: Sum from 1 to N

Steps

1. In plain language:

- What is "sum from 1 to N"? $\frac{n(n+1)}{2}$

2. Write the algorithm:

- Choose a value for n, then add 1 to n.
- then multiply n with the added number
- then divide the resultant by 2
- then the value derived is the sum of numbers from 1 to n

3. Now code it:

algorithm

python

```
N = int(input("Enter N: "))
```

```
total = 0
```

```
for i in range(1, N+1):
```

```
    total = total +  $\frac{N(N+1)}{2}$ 
```

```
print("Sum =", total)
```

1. enter the value of N

2. enter the range as (1, N+1)

3. use the formula $[N(N+1)]/2$

4. Run the program.

4. Try N = 5, 10, 100 and check results.

$$(5) = 15$$

$$(10) = 55$$

$$(100) = 5050$$

```
N = int(input("Enter N: "))
```

```
total = 0
```

```
for i in range(1, N+1):
```

```
    total = total + i
```

```
print("Sum =", total)
```

Activity 4.2 – Star Patterns

Title: Loops for Patterns

Pattern (triangle)

```
*  
**  
***  
****
```

1. Explain the pattern in words:

an asterisk "*" is added to a new row, for the mentioned range

2. Write code:

python

```
rows = int(input("Enter number of rows: "))
```

```
for i in range(1, rows + 1):
```

```
    print("*" * i)
```

3. Try different rows and observe.

rows = int(input("Enter the rows: "))

for i in range(1, rows + 1):

print(" " * (rows - i) + "*" * i)



3 2 1



Activity 5.1 – Working with Lists

Title: Analyze Marks List

Given

python

marks = [56, 78, 90, 45, 88]

1. In words, write how to find:

- Maximum mark: max()
- Average mark: sum() / len()
- Count of marks ≥ 60 : check each marks against the condition

2. Now write Python:

python

marks = [56, 78, 90, 45, 88]

Max

max_mark = max(marks)

Average

total = 0

for m in marks:

total = sum(marks)

average = total / len(marks)

Count ≥ 60

count = 0

for m in marks:

if m >= 60:

count = count + 1

print("Max:", max_mark)

print("Average:", average)

print("Count ≥ 60 :", count)

Activity 5.2 – Choosing an Approach

Title: Which Method is Better?

Problem:

Given a list of 1000 numbers, find how many are positive.

1. Approach A (loop) – describe in words:

loop runs the iteration through all 1000 nos.

2. Approach B (using built-in functions or list comprehension) – describe:

using sum() or len() function

3. Which is easier to understand for you now? Why?

loop.

4. Implement one approach in Python:

python

numbers = [10, -3, 5, 0, 8, -2] # example

Your code

count = 0

for in numbers:

if num > 0:

count += 1

print ("The no. of positive no. is:", count)

Activity 6.1 – Creating a Function

Title: Wrap Logic into a Function

Task

Turn average-of-marks logic into a function.

1. In words: what does the function do?

this function takes the data & turns it into avg.

2. Write code:

python

```
def average_marks(marks_list):
```

```
    total = 0
```

```
    for m in marks_list:
```

```
        total = total + m
```

```
    avg = total / len(marks_list)
```

```
    return avg
```

} none out of the func loop.

```
data = [60, 70, 80]
```

```
print("Average =", average_marks(data))
```

3. Test with another list and note output.

(100, 90, 50) ⇒ 80.0

Activity 6.2 – Problem-Solving Template

Title: Standard Template for Any Problem

Use this for any new problem. → base fee, age below 12 yr - 50% off & age above 60 yr - 30% off or no discount

1. Problem understanding

- What is given (inputs)? → base fee & age.

(discount for ages below 12 is 50% & above 60 is 30%)

- What is required (outputs)?

to calculate the final fee after discount or without discount based on age

- Any constraints (limits, ranges)?

— discount for age below 12 is 50% & above 60 is 30% off or there is no discount.

2. Analysis

- Write steps in bullets (algorithm):
 - Set the base fee
 - Check the age, if it is below 12 or above 60 or b/w them
 - Calculate the discount & final fee, if it satisfies the condition.
 - Print the final fee for that age.

3. Design

- Will you use loops? Which? ✓
- Will you use conditions (if/else)? ✓
- Will you use functions? Which ones? ✓
- What data structures (list, etc.)? ✓

4. Code (Python)

python

Your implementation

5. Test

- Test case 1: input 11 → output 50
- Test case 2: input 75 → output 70
- Edge case: input 21 → output 100

Test case 1: i/p ⇒ base fee = 100
age = 11 o/p ⇒ final fee = 50.

Test case 2: i/p ⇒ base fee = 100
age = 75 o/p ⇒ final fee = 70

Test case 3: i/p ⇒ base fee = 100
age = 21 o/p ⇒ final fee = 100

y) a) Base fee & Age
b) Final fee (After discount)
c) Discount: Age < 12 — 50% off
Age > 60 — 30% off
If age b/w 12 & 60 — no discount

if age b/w 12 & 60 - no discount

A class charges a base fee.

- If student is below 12 years, 50% discount.
- If above 60, 30% discount.
- Otherwise, no discount.

(Attach a blank copy of the template and ask students to fill all five sections.)

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```
base_fee = int(input("Enter base fee: "))
discount_age = int(input("Enter age: "))
```

if age < 12 :

$$\text{discount} = 0.05$$

elif age > 60:

$$\text{dis count} = 0.03$$

else :

des comit = 2

$$\text{final_fee} = \text{disc_fee} * (1 - \text{discmt})$$

```
print ("final fee is {final-fee})"
```

- 2) * Enter base fee & age
 - * Check if age satisfies the given condition
 - * Calculate the discounted fee & final fee after calculations
 - * Print the final fee
- 3) b) Yes (if-elif-else)

3) b) Yes (if-elif-else)

Activity 7.2 – Reflection Sheet

Title: How I Solve a New Problem

1. When I see a new problem, my first step is:

Identify problems, Check the given conditions

2. Before coding, I make sure I know:

- Inputs: values we need to input/predefine/declare
- Outputs: what output I need after computation/ calculation

3. The tools I think about (loops, if, functions) are:

For loop, if - elif - else, nested loops, functions.

4. One thing I will do differently from now on:

give proper indentations, use proper loops & make the code complete in lesser steps after understanding it.